

PUBLIC SUBMISSION

As of: March 21, 2025
Received: March 15, 2025
Status: [REDACTED]
Tracking No. m8a-up0m-icqs
Comments Due: March 15, 2025
Submission Type: API

Docket: NSF_FRDOC_0001

Recently Posted NSF Rules and Notices.

Comment On: NSF_FRDOC_0001-3479

Request for Information: Development of an Artificial Intelligence Action Plan

Document: NSF_FRDOC_0001-DRAFT-3479

Comment on FR Doc # 2025-02305

Submitter Information

Email: [REDACTED]

Organization: Artificial Intelligence Infrastructure Coalition (AIC)

General Comment

On behalf of the AI Infrastructure Coalition, CO2EFFICIENT is pleased to submit a response to the Request for Information regarding an AI Action Plan from the National Coordination Office (NCO) of the Networking and Information Technology Research and Development (NITRD) and the National Science Foundation.

We hope this information is useful in the development of an AI Action Plan to advance America's AI leadership. Please feel free to reach out if you have any questions or would like any elaborations.

Thank you for the opportunity to contribute to this priority work.

See attached file(s)

Attachments

AI Infrastructure Coalition Response to NSF RFI 250315



March 14, 2025

Attn: Faisal D'Souza, NCO
AI Action Plan
2415 Eisenhower Avenue
Alexandria, VA 22314
Via email: [REDACTED]

Re: Request for Information on the Development of an Artificial Intelligence (AI) Action Plan

To Whom It May Concern:

As representatives of the digital infrastructure ecosystem, members of the AI Infrastructure Coalition (AIC) appreciate the opportunity to provide comments on the [Request for Information \(RFI\) on the AI Action Plan](#) issued by the Networking and Information Technology Research and Development (NITRD) National Coordination Office (NCO), National Science Foundation on behalf of the White House Office of Science and Technology Policy (OSTP) on February 6, 2025. We understand that this RFI is intended to collect input for creating an AI Action Plan to advance America's AI leadership as called for under [Executive Order 14179](#), "Removing Barriers to American Leadership in Artificial Intelligence" in order "to establish a U.S. policy for sustaining and enhancing America's AI dominance in order to promote human flourishing, economic competitiveness, and national security."

Pursuant to the RFI and EO 14179, on behalf of the Artificial Intelligence Infrastructure Coalition (AIC), this submission provides recommendations regarding the need for robust and modernized infrastructure that can support AI, enable economic growth, and enhance the resilience of the U.S. digital infrastructure ecosystem.

The growth of artificial intelligence (AI) is a transformative force, shaping the economic, technological, and industrial landscape in the U.S. Looking ahead, there is an opportunity to foster the resurgence of American manufacturing and cement the U.S. as the leader of the digital economy, but, there is a simultaneous need for additional focus on the infrastructure needs that are critical to enabling the full deployment of AI technologies.

Unfortunately, the U.S.' current electrical infrastructure is not fit for purpose to drive our digital economy. Our grids and wires are outdated and not designed for the demands of the AI digital age. The current state of the country's electricity grid—characterized by underinvestment, aging infrastructure, and lengthy time to permit and build new lines—poses a significant barrier to growing the U.S.' digital economy and scaling AI deployment.

Furthermore, many of the U.S.' existing data centers are not optimized for accelerated computing capabilities including AI, high-performance computing (HPC), and Quantum computing. (**Note:** for the purposes of these comments, all categories of accelerated computing are categorized as "AI computing"). AI computing requires more power and cooling than traditional computing.

The comments we are providing focus on the need for robust and modernized infrastructure that can support AI, enable economic growth, and enhance the resilience of the U.S. digital infrastructure ecosystem. It is critical that the U.S. shore up its physical infrastructure both in terms of the electric grid and data centers' power and cooling capabilities to enjoy the full benefits of AI's applications and enable the Trump Administration to achieve its goal of "sustaining and enhancing AI's dominance."

Defining America's Digital Infrastructure Ecosystem

AI computing requires energy-intensive technologies that depend on modernized electricity grids, specialized processors, fit for purpose cooling systems, and high-density data centers. The undersigned signatory represent the digital infrastructure ecosystem in the U.S.

The Artificial Intelligence Infrastructure Coalition (AIC) is a coalition comprised of members that span the entire AI ecosystem (i.e., AI platform providers, microchips, server hardware, data center equipment, software developers, data center operators, and power management services) that address the dynamic and changing needs of the digital infrastructure industry. At the heart of this ecosystem are AI chip manufacturers, which design specialized chips (e.g., graphical processing units (GPUs) and AI accelerators), optimized for processing large AI workloads. Chips are integrated into servers by server original equipment manufacturers (OEMs), which tailor their hardware to meet the performance demands of AI applications. Data center equipment providers offer additional infrastructure, such as networking components, high-performance storage, and cooling solutions, to ensure optimal performance. Equally vital are backup power and power management systems, which ensure uninterrupted operation during power failures or grid fluctuations, a critical factor in maintaining uptime for AI operations. Data center designers and operators play the role of architects, constructing and managing facilities that can house these power-intensive technologies, ensuring optimal layout, airflow, and energy efficiency.

AIC represents key aspects of the digital infrastructure ecosystem. All the various policies they promote are critical to ensuring the U.S. digital infrastructure functions harmoniously to meet the growing needs of AI development and deployment.

Data Centers

Data centers are the foundation of the digital infrastructure on which our modern society and economy are built. Data centers are physical locations that organizations use to house their critical applications and data. Anything that takes place online “in the cloud” is powered by a data center. Data centers aggregate our collective computing demands efficiently and securely in one location.

Inside the data center are routers, switches, firewalls, storage systems, servers, and application-delivery controllers. The digital infrastructure is supported by power subsystems, uninterruptible power supplies, ventilation, cooling systems, backup generators, and network connections.¹ The technologies supporting data centers, and the power grids on which they rely, are dependent on a robust supply chain of components. The acquisition process for transformers, breakers, transmission lines, grid connections, etc. needs to be resilient and reliable across a variable global market to avoid capacity bottlenecks and service interruptions.

The complex component and capacity needs of AI computing have driven data centers to evolve far beyond traditional client-server models, pushing new paradigms in infrastructure design, resource allocation, power management, and operational strategy.

The Electricity Grid and Power Generation

The growth of data centers in recent years to support the growing economy and AI has made clear the interdependence between digital infrastructure and the electricity grid. The growth of digital infrastructure, particularly AI, is impossible without corresponding investments in the electric grid. However, after two decades of flat electricity demand, the U.S. grid is struggling to accommodate the resurgence of load growth driven by the expansion of data centers, the reshoring and resurgence of manufacturing, semiconductor fabrication, and transportation and building electrification.² Grid planners are being caught off guard by the rapid increase in load growth, which is expected to nearly double over the next five years.³ Infrastructure planning required little attention when the U.S. had no growth. However, that is no longer the case.

¹ “About the Data Center Industry.” Data Center Coalition. <https://www.datacentercoalition.org/data-center-industry>.

² Wilson, John D., and Zach Zimmerman. “The Era of Flat Power Demand is Over.” *Grid Strategies*, December 2023, p. 3. <https://gridstrategiesllc.com/wp-content/uploads/2023/12/National-Load-Growth-Report-2023.pdf>.

³ Ibid.

The U.S. digital infrastructure ecosystem faces multiple challenges, ranging from regulatory obstacles to grid limitations. However, the biggest challenge facing the U.S. digital infrastructure ecosystem is the strained electricity grid. The current electricity grid, which was designed to serve the needs of an industrial economy, is ill-equipped to handle the demands of a digital economy powered by AI.

Today, our strained electric grid infrastructure and our inability to expeditiously build new transmission and generation seems poised to undermine our ability to bring an American economic renaissance, one spurred by the digital economy and scaling AI development, to bear. The U.S.' existing electricity infrastructure is stymying the digital infrastructure ecosystem. The policies and permitting requirements that govern the processes for upgrading our electricity infrastructure and the processes to bring new generation resources online are woefully inadequate and cumbersome to handle new growth reliably and cost-effectively.

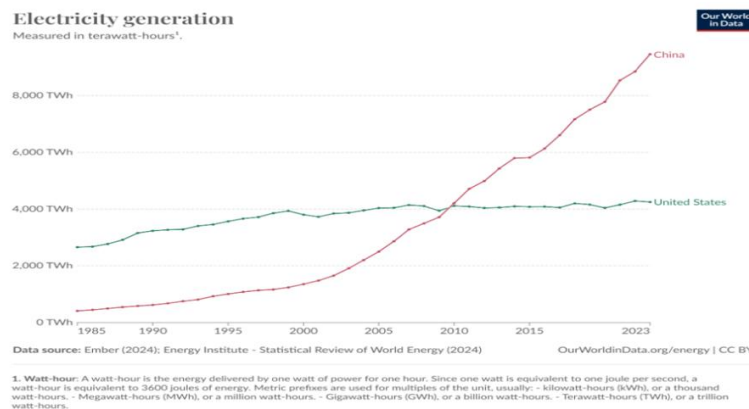
Addressing the inadequacies of the U.S. electricity grid and working with federal agencies and states through Public Utility Commissions (PUCs) and energy offices to support AI computing offers a unique opportunity to improve the grid for all customer classes. Such updates will not only benefit technology companies and research institutions but will also have far-reaching positive impacts on manufacturing, commercial businesses, and residential customers.

Global Competition: Ensuring that AI Development Remains in the U.S.

The global economy is facing a changing paradigm. AI has become a critical technology that will shape the future of global competitiveness, national security, and economic prosperity. This presents an opportunity for the U.S., home to many of the companies driving this transformation, to take the lead. And, through [EO 14179](#), the Trump Administration has made clear its priority of "sustaining and enhancing America's global AI dominance." This was the impetus for President Trump's January 23, 2025 announcement of the Stargate project, a \$500 billion private sector effort to build new AI data centers across the country.

Put simply, the U.S. is in a race to compete with our rivals, particularly China, on the development and deployment of AI and its corresponding electric infrastructure. We need a national sense of urgency to establish the electric infrastructure that can power the U.S.' digital economy. While the U.S. struggles to meet the demand of new load growth, China continues to rapidly build electricity generation capacity and transmission infrastructure while developing its own AI capabilities. "Twenty years ago, Chinese energy consumption was a fraction of U.S. levels and not much higher than Germany. Today, China's primary energy and

power consumption are larger than the U.S. and Germany combined.”⁴ For example, in 2022, China spent over \$165 billion on its electricity grids while the U.S. spent less than \$33 billion.⁵



Reliable and affordable electricity is vital to drive the economic engine of AI and the U.S.’ growing digital economy. However, electric power regulation has become an obstacle rather than an enabler of efficient and reliable operation of the grid and its necessary modernization. There must be a faster and more coherent way of addressing permitting and multi-state infrastructure development to enable this economic engine to grow and compete with our rivals, particularly China. Specifically, to achieve the Administration’s goal of global AI dominance, the U.S. must ensure that this country is able to compete with foreign countries on the following four key factors that are necessary for scaling AI development and leadership in the U.S.

1. Accessibility of power: Many foreign countries or regions offer cheaper electricity rates and/or quicker access to power that could be more attractive to data center developers. As power consumption continues to increase, the cost of electricity becomes a significant factor in determining the feasibility of new data center sites.
2. Government incentives: Many foreign markets offer substantial tax breaks, subsidies, and other incentives to attract data center investments. These incentives can significantly reduce capital expenditure and operational costs, making certain regions more appealing than the U.S., if equivalent support is not provided domestically.
3. Permitting and regulatory barriers: In some foreign markets, permitting and regulatory hurdles are streamlined, allowing faster development of data centers. Lengthy permitting processes or restrictive regulations in the U.S. can slow down projects, making international locations with more business-friendly environments preferable for data center expansion.

⁴ Cembalest, Michael. “Electravisión.” *J.P. Morgan 14th Annual Energy Paper*, March 2024, p. 16. <https://assets.jpmprivatebank.com/content/dam/jpm-pb-aem/global/en/documents/eotm/electravisión.pdf>.

⁵ Ferris, Nick. “Weekly Data: Grid Investment in China More Than Every Other Country Combined.” *Energy Monitor*, March 15, 2024. <https://www.energymonitor.ai/tech/networks-grids/weekly-data-grid-investment-in-china-more-than-every-other-country-combined/>.

4. Access to talent: Workforce availability is another critical factor. The shortage of skilled professionals, such as network engineers and cybersecurity experts, can drive companies to seek out markets where there is a more readily available talent pool to meet their needs.

Deploying AI to Improve the Grid

AI has the potential to dramatically change how the U.S. manages its electricity grid. By leveraging AI for grid optimization, utilities can improve the efficiency of electricity generation and distribution, reduce outages, and better integrate renewable energy sources.⁶ For example, AI can play a vital role in grid operation and optimization by helping network operators understand grid conditions in real time, make more informed decisions, and forecast potential grid disruptions.⁷ AI systems can draw huge amounts of data from a range of sources to aid in planning (e.g., which grid equipment to put in which locations), identifying grid infrastructure that needs to be reinforced to deal with extreme weather, monitoring grid performance to improve system efficiency and stability, and providing real-time assessments of the maximum amount of power that transmission lines can carry (i.e., dynamic line rating).⁸ AI can support enhanced grid maintenance by facilitating remote inspections of infrastructure, identifying operating parameters that will extend equipment lifetimes, and pinpointing early data trends to predict future failures and arrange for preventive maintenance of equipment.⁹ AI can also greatly enhance overall electricity usage and capacity of the grid through supporting the development of “virtual power plants” that can provide grid services and demand response.¹⁰

Relatedly, building operators are deploying AI-driven power management systems to optimize energy distribution, monitor workloads, and adjust cooling in real time. This dynamic management can reduce energy consumption by precisely matching energy usage to current computational needs.¹¹ The Electric Power Research Institute (EPRI) recently launched its DCFlex Initiative, which includes both data center companies and utilities, and is exploring

⁶ See U.S. Department of Energy, AI for Energy: Opportunities for a Modern Grid and Clean Energy Economy, April 2024, https://www.energy.gov/sites/default/files/2024-04/AI%20EO%20Report%20Section%205.2g%28i%29_043024.pdf; see also Rozite, Vida, Jack Miller, and Sungjin Oh, “Why AI and Energy Are the New Power Couple – Analysis,” International Energy Agency, November 2, 2023, <https://www.iea.org/commentaries/why-ai-and-energy-are-the-new-power-couple>; Dannouni, Amame, et al., Accelerating Climate Action with AI, November 2023, <https://web-assets.bcg.com/72/cf/b609ac3d4ac6829bae6fa88b8329/bcg-accelerating-climate-action-with-ai-nov-2023-rev.pdf>.

⁷ Kim, June. “Four Ways AI Is Making the Power Grid Faster and More Resilient.” *MIT Technology Review*, November 22, 2023. <https://www.technologyreview.com/2023/11/22/1083792/%20ai-power-grid-improvement/>.

⁸ Sandalow, David, et al. “Artificial Intelligence for Climate Change Mitigation.” *Innovation for Cool Earth Forum*, December 2023, p. 39. <https://www.icef.go.jp/wp-content/uploads/2024/02/AI-Climate-Roadmap-ICEF-Dec-1-2023.pdf>.

⁹ World Economic Forum. “Harnessing Artificial Intelligence to Accelerate the Energy Transition.” September 2021, p. 10-11. https://www3.weforum.org/docs/WEF_Harnessing_AI_to_accelerate_the_Energy_Transition_2021.pdf; Renshaw, Jeremy. “The Role of Artificial Intelligence in Powering America’s Energy Future.” *Statement before the House Committee on Energy and Commerce, Subcommittee on Energy, Climate, and Grid Security*, 118th Cong., October 19, 2023, p. 5. https://d1dth6e84htgma.cloudfront.net/10_19_23_ENG_Testimony_Renshaw_931baf42d5.pdf; Kim, “Four Ways AI Is Making the Power Grid Faster.”

¹⁰ Downing, Jennifer, et al. “Pathways to Commercial Liftoff: Virtual Power Plants.” *U.S. Department of Energy*, September 2023, Appendix VII. Potential Grid Services. https://liftoff.energy.gov/wp-content/uploads/2023/10/LIFTOFF_DOE_VVP_10062023_v4.pdf.

¹¹ See Intel. “Power Management – Leveraging AI for Smarter Data Center Power Efficiency Solution Brief.” *Intel Network Builders*, September 2022. <https://networkbuilders.intel.com/docs/networkbuilders/power-management-leveraging-ai-for-smarter-data-center-power-efficiency-solution-brief-1664790296.pdf>; Dafinger, Klaus. “Meeting Data Center Cooling Demands in the AI Era.” *Data Center Dynamics*, July 10, 2024. <https://www.datacenterdynamics.com/en/opinions/meeting-data-center-cooling-demands-in-the-ai-era/>.

innovative strategies to help enable operational and deployment flexibility, streamline grid integration, and transition backup power solutions to grid assets.¹²

With respect to renewable power generation, AI can help with siting and sizing large solar and wind projects, maximizing production, and predicting and scheduling maintenance to minimize failures. AI can also improve predictions of supply and demand.¹³ As more digitized, distributed energy resources are added to the grid, AI will play an important role in managing energy demand and distributed resources for better integration and enhanced grid reliability.¹⁴

Recommendations for the AI Action Plan to Improve the Digital Infrastructure Ecosystem in the U.S.

Representatives across the digital infrastructure ecosystem stand ready to work with policymakers, utilities, and other stakeholders to ensure that we can provide the necessary infrastructure to power America's growing digital economy. Below are recommendations to include in the AI Action Plan:

1. Modernizing the Grid

- We recommend the AI Action Plan adopt policies to enable the U.S. infrastructure improvements required to support AI computing. The plan needs to address not only adding more power but also making the grid more efficient, resilient, and capable of supporting a range of modern technologies. Grid resilience improvements would protect manufacturing plants and homes from blackouts and extreme weather disruptions, benefiting every customer class. Below are specific recommendations for modernizing the grid.
 - i. *Develop permitting reforms:* The U.S. faces a significant challenge with its strained electricity grid infrastructure and the slow pace of building new transmission and generation capacity. Electric power regulation has become an obstacle, rather than an enabler of efficient and reliable grid modernization. A faster, more streamlined approach to permitting and multi-state infrastructure development is needed to drive economic growth and compete globally, particularly with China. There is broad, bipartisan agreement that unrestricted judicial review delays energy projects and related infrastructure. We recommend that the AI Action Plan propose permitting reforms to ensure greater certainty for energy projects while maintaining crucial environmental and community protections that help grow community support for data centers in the localities where they reside.
 - ii. *Explore additional interconnection reforms:* In July 2023, the Federal Energy Regulatory Commission (FERC) issued a unanimous, bipartisan decision, Order

¹² Electric Power Research Institute. "DCFlex: Optimize Data Center Operational Flexibility to Help Strengthen the Grid." EPRI, 2025. <https://publicdownload.epri.com/PublicAttachmentDownload.svc/AttachmentId=90809>.

¹³ Lu, W., J. Wang, X. Chen, and Y. He. "A Data-Driven Method for Predicting Failures in Solar Panels and Wind Turbines." *IEEE Transactions on Sustainable Energy* 9, no. 2 (2018): 734-743.

¹⁴ IRENA. *Artificial Intelligence and Big Data for Energy Systems*. Abu Dhabi: International Renewable Energy Agency, 2022, p. 9-10, <https://www.irena.org/publications/2019/Sep/Artificial-Intelligence-and-Big-Data..>

- 2023, to improve generator interconnection processes.¹⁵ The key reform is a requirement for grid operators to adopt a cluster-study approach, allowing interconnection requests to be reviewed in groups rather than individually. Order 2023 also includes secondary reforms, such as improved transparency of the transmission system. While these changes are beneficial, they will not significantly alter the current challenges with interconnection.¹⁶ The AI Action Plan should include a consultation with stakeholders to determine what additional interconnection reforms could help to unlock incentivizing utilities to add new generation capacity.
- iii. *Ensure robust implementation of transmission reforms:* Consumers benefit when independent economic planning replaces inefficient utility planning. As revised, FERC Order 1920-A promotes planning over an appropriate time horizon, requires scenario analysis to mitigate risk, and uses higher-quality cost-benefit analysis, which was intended to address the negative repercussions of reactive, piecemeal approaches.¹⁷ We recommend that the AI Action Plan study how proper implementation of Order 1920, coupled with a strong state role in scenario development, cost allocation, and benefit considerations, could help speed approvals for new essential cost-effective electric infrastructure to support growing demand growth.
 - iv. *Ensure co-location of new supply and new load remains an option:* The issue of co-locating large loads at power plants highlights the need for wholesale rules that minimize system costs and allocates them based on customers' benefit. Recently, FERC [voted unanimously](#) to launch a review of issues associated with the co-location of large loads such as AI-enabled data centers at generating facilities in the PJM Interconnection. When done correctly, all customer classes can benefit. Policymakers and regulators should ensure that wholesale tariffs reflect cost based on customer's benefits. For example, the Administration should encourage FERC to finalize how network costs are recovered to ensure fairness and create opportunities for its customers to avoid future investments, beyond just meeting utility revenue requirements.¹⁸
 - v. *Incentivize Data Centers to Select Locations with Available Capacity:* We recommend the AI Action Plan explore opportunities to reform electricity market designs to incent demand side innovation by rewarding placing demand where the electric grid has available capacity. For example, high wind-producing areas¹⁹ often have surpluses of local, air pollution-free generation

¹⁵ Federal Energy Regulatory Commission. "Order No. 2023: Improvements to Generator Interconnection Procedures and Agreements." July 28, 2023. <https://www.ferc.gov/media/order-no-2023>.

¹⁶ Hartman, Devin and Beth Garza, "Finishing Generator Interconnection Reform." R Street, March 3, 2023. <https://www.rstreet.org/commentary/finishing-generator-interconnection-reform/>

¹⁷ Federal Energy Regulatory Commission. "Order No. 1920: Building for the Future through Electric Regional Transmission Planning and Cost Allocation." May 13, 2024. <https://www.ferc.gov/media/e1-rm21-17-000>.

¹⁸ Electricity Customer Alliance, et al., "Re: Large Loads Co-located at Generating Facilities," Docket No. AD24-11-000, submitted via eFiling to the Federal Energy Regulatory Commission, October 30, 2024. https://elibrary.ferc.gov/eLibrary/filelist?accession_number=20241030-5112&optimized=false.

¹⁹ Burgum, Doug. "One of World's Largest Data Centers to Locate in Williston Area as Industry Targets Growth in ND." North Dakota Office of the Governor, January 26, 2022. <https://www.governor.nd.gov/news/burgum-one-worlds-largest-data-centers-locate-williston-area-industry-targets-growth-nd>.

- that attract data center development. These practices reduce infrastructure strain and offer clean per-unit emissions from data center expansion. This is why large consumers, including data centers, are advocating for reforms²⁰ to provide direct market access in regions with excess available capacity.²¹
- vi. *Scaling new Collaborative Tariff Structures*: The AI Action Plan should include plans for encouraging public utility commissions (PUCs) to implement scaled tariff structures. Utilities and large load customers, such as data center operators, are collaborating to develop new tariff structures that support the deployment of advanced nuclear energy, long-duration energy storage, and other technologies. This collaboration highlights the innovation possible through joint planning between utilities and customers. The federal government should explore ways to facilitate this type of collaboration and scale utility tariff structures that promote new business models and innovation, specifically focusing on ensuring consistency across regions.²²
 - vii. *Facilitate State Economic and Electricity Planning Processes*: The AI Action Plan should promote state and PUC economic development coordination. Establishing realistic load forecasts and clear policies that protect customers is crucial for utilities to meet their service obligations. As electric load growth accelerates across parts of the country, states and PUC commissioners should ensure each state PUC has transparent and non-discriminatory policies in place to respond to continued unforeseen changes in economic growth, demonstrated via electric demand.
 - viii. *Eliminate Barriers to Optimizing Existing Transmission Infrastructure*: The AI Action Plan should prioritize grid enhancing technologies (GETs) to optimize the existing transmission system. Despite growing awareness of opportunities to incentivize market-driven technologies that optimize the existing transmission system, barriers persist in both cost-of-service and non-cost-of-service states. Additional processes should be established to improve visibility and promote technologies, such as GETs, that reduce transmission grid congestion. For example, California Governor Gavin Newsom recently signed SB 1006, which requires utilities to evaluate GETs, such as dynamic line ratings, advanced power flow control devices, and topology optimization, at least every two years as part of transmission planning.²³
 - ix. *Establish Leadership on Proactive Transmission Planning*: The AI Action Plan should include collaboration with states to promote the establishment of more robust transmission planning processes. FERC's Order 1920-A aims to enhance transmission planning practices and, when properly implemented, will improve

²⁰ Villarreal, Chris. "Comments of the R Street Institute Before the Iowa Utilities Board on Policy Charrette #2." *R Street Institute*, October 4, 2023. <https://www.rstreet.org/outreach/comments-of-the-r-street-institute-before-the-iowa-utilities-board-on-policy-charrette-2/>.

²¹ Hartman, Devin. "Commercial & Industrial Consumer Market Access." *R Street Institute*, June 12, 2024. <https://www.rstreet.org/wp-content/uploads/2024/07/Hartman-ESIG-June-2024-CI-Market-Access.pdf>.

²² Yahoo Finance. "Duke Energy Responding to Growing Demand with Solar and Energy Storage Projects." October 17, 2023. <https://finance.yahoo.com/news/responding-growing-demand-duke-energy-150000676.html>.

²³ Switchgear Magazine. "California Governor Signs Bill Requiring Utilities to Consider Grid-Enhancing Technologies." October 11, 2023. <https://switchgear-magazine.com/tm-news/business/california-governor-signs-bill-requiring-utilities-to-consider-grid-enhancing-technologies/>.

processes without Regional Transmission Organizations as well.²⁴ Consistent implementation will be crucial to ensure that customers benefit from improved transmission. Additional reforms will also be necessary to incorporate interregional transmission planning processes that support the cost-effective development of long-range, high-voltage transmission.²⁵

- x. *Unleash Demand-side Innovation:* The AI Action Plan should include a proposal to work with states to enable demand-side innovations. As new loads seek to implement their own power solutions to support excess generation capacity, states must establish processes that promote demand flexibility, on-site generation development, and incentives to locate demand where it places the least burden on existing infrastructure. Responding to growing energy volatility will require decisive action not only to secure and maintain ample energy supply but also to leverage innovative technologies to measure, forecast, and optimize energy demand. A comprehensive energy strategy will respond to both supply-side and demand-side contributions to energy volatility. As of 2020, more than 60% of energy used for electricity generation in the U.S. was lost during conversion.²⁶ To foster demand-side solutions, state economic development agencies should play a larger role in understanding electric infrastructure needs and challenges. Additionally, stakeholders must be involved in the planning process with greater transparency and a new level of collaboration. The data center and AI boom represents a bold opportunity to leverage a key segment of America's economic engine to stabilize and power the grid. By making the most of our new and existing energy resources – including via growing markets and segments (inclusive of data centers, re-shored manufacturing, etc.) – the Administration can power America's energy independence and ensure America leads the world in energy affordability, innovation, and reliability.

2. *Research and Development:*

- We recommend that the AI Action Plan create a special program in one or more of the DOE's National Laboratories (National Lab) or establish a collaboration with a laboratory at an existing university to evaluate compute resource allocation strategies for large-scale foundation-model training and/or other AI use cases. This program should focus on public-private collaboration, with DOE playing role in co-designing AI solutions that address public sector needs, including meeting power and energy demands sustainably. Such a program should be designed to ensure that, while the private sector is focused on commercialization, DOE prioritizes AI applications directed towards societal benefits. Some of the critical topics for AI computing that we recommend be included in such a research program are:
 - i. Small AI model development and efficient software design.

²⁴ Federal Energy Regulatory Commission. "Order No. 1920: Building for the Future through Electric Regional Transmission Planning and Cost Allocation." May 13, 2024. <https://www.ferc.gov/media/e1-rm21-17-000>.

²⁵ Hartman, Devin, and Kent Chandler. "Eliminating the Discord Over FERC Order 1920: What Role for States?" *R Street Institute*, July 2, 2024. <https://www.rstreet.org/commentary/eliminating-the-discord-over-ferc-order-1920-what-role-for-states/>.

²⁶ U.S. Energy Information Administration. "Global Crude Oil Demand Is Expected to Increase in 2021 after Falling in 2020." *Today in Energy*, 2025. <https://www.eia.gov/todayinenergy/detail.php?id=44436>.

- ii. Compilation of current cooling systems capable of managing thermal densities of 1500kw+ rack densities.
 - iii. A compilation and assessment of current research on reliability and compatibility with electronics equipment of various liquid cooling fluids.
 - iv. Feasibility of heat reuse in data centers in the U.S.
3. *Create a Working Group to Identify and Address Challenges Confronting the Data Center Industry:*
- We recommend that the AI Action Plan establish a working group with industry and other stakeholder groups to identify and address challenges confronting the data center industry. The working group should meet quarterly and contain representatives from across the digital infrastructure ecosystem, including power providers, data center developers, grid operators, and federal and state policymakers to identify and address policy hurdles to designing energy-efficient, resilient infrastructure that supports the growth of AI while minimizing resource use and enhancing reliable operations.
4. *Create a Dedicated AI “Policy Home” within the Executive Branch:*
- While federal agencies, including the Department of Energy, the General Services Administration, and the White House Office of Science and Technology Policy, have taken a range of actions related to AI, there is currently no coordinated federal approach to data center efficiency and use. As the demand for data centers continues to grow, it is crucial to prioritize efficient and sustainable operations.
5. *Create AI Development Zones:*
- We recommend that the AI Action Plan include plans for developing a comprehensive map of areas across the U.S. that could support future data center development. These areas could be designated as AI Development Zones, where permitting and siting restrictions are eased to facilitate the construction of large-scale data centers and localized power generation. These zones would provide long-term economic development opportunities in regions that house them. It is important to ensure that AI Development Zones include the following features:
 - i. Ability to develop redundant fiber paths and support right-of-way access for further development, if needed.
 - ii. Availability of highly reliable, affordable electricity.
 - iii. A strong, shovel ready site, access to construction labor, and reasonable construction costs.
 - iv. Proximity to technical engineering talent and local universities.
 - v. Clear and efficient state and local approval processes necessary for development, construction, and operations.
 - vi. Favorable local and state tax and fiscal policies plus a welcoming local community with a business-friendly climate and strong partnerships.

Conclusion

The U.S. stands at a crossroads. While AI development is crucial, the infrastructure supporting it—particularly electricity grids and data center systems—must not be overlooked. As AI computing continues to drive economic growth and innovation, the country must also invest in the infrastructure needed to support these advancements. As it stands, the rapid growth of AI computing has brought the inadequacy of the U.S. electricity infrastructure into sharp focus. AI holds tremendous potential for improving productivity, optimizing energy use, and transforming industries, however, its success will depend on the strength and resilience of the U.S. electricity grid and the ability of our data centers to provide adequate power and cooling for AI computing.

Without significant investment in grid modernization, the U.S. risks falling behind in the global race for AI leadership. Policymakers must act urgently to address the challenges facing the U.S. digital infrastructure ecosystem, including aging electricity infrastructure, regulatory obstacles, and growing energy demands. By prioritizing investments in transmission, distribution, and generation, the U.S. can ensure that it remains at the forefront of AI development and that the benefits of this technological revolution are felt across all sectors of the economy.

Now is the time for serious conversations among *all* policymakers across the federal government – including the White House, the DOE, the Department of Commerce, FERC, and Congress – about how to speed up the pace to build electricity infrastructure and systems that will power the digital and new AI economy for years to come. The AIC is happy to coordinate industry stakeholders to inform the discussion.

Respectfully submitted,

AI Infrastructure Coalition